

I Year II Semester

PHYSICS FOR ADVANCED COMPUTING (CSE & Allied Branches)

(AI Integrated)

Course Code	Year/Sem	L	T	Pr	P	Credits
26APH05T	I Year II Sem	3	0	0	0	3

Prerequisites: Engineering Mathematics, Basic Electronics, Computer Science Fundamentals

Course Objectives

The objective of this course is to:

1. To equip students the concepts of interference, diffraction and polarization and their applications in optics and communication systems with basic AI-assisted analysis.
2. To introduce lasers, fiber optics and advanced sensing technologies along with simple AI-assisted signal processing techniques.
3. To provide basic knowledge of quantum mechanics and quantum computing with simple AI-assisted quantum applications.
4. To familiarize students with semiconductors and magnetic materials and their applications using basic AI-assisted material analysis methods.
5. To impart knowledge on low temperature physics and superconductivity and their applications with simple AI-assisted monitoring techniques.

Course Content

UNIT I: Wave Optics and Quantum Information Processing 9hrs

Core Content:

Interference: Principle of superposition and coherence, Interference, Types of interference, Interference in thin films (reflection geometry) and Newton's rings. Applications: Determination of wavelength and refractive index.

Diffraction: Introduction to diffraction, Fresnel and Fraunhofer diffraction, Fraunhofer diffraction due to single, double and N-slits (Diffraction Grating), Dispersive power and Resolving power of grating. Applications: CD/DVD data storage, Barcode scanners.

Polarization: Types of polarization, Nicol's prism, Half-wave and Quarter-wave plates.

AI Integration:

AI-assisted polarization management for quantum key distribution. AI-integrated holographic interferometry for surface flatness analysis.

Free/Open-Source AI Tools: Python (NumPy, SciPy, Matplotlib and scikit-learn).

UNIT II: Quantum Photonics and Advanced Sensing 9 hrs

Core Content:

Lasers: Interaction of radiation with matter, Requisites and conditions for laser action, He-Ne laser, Nd-YAG laser, LiDAR basics. Laser-based data communication, Free-space optical communication, Optical data storage, Photonic computing basics, Applications of lasers.

Fiber Optics: Propagation mechanism, Numerical aperture, Acceptance angle, Types of optical fibres, Classifications and Modes of optical fibres, Attenuation and Losses in optical fibers, V- Number - Distributed Fiber Optic Sensing (DFOS) with AI-Powered Signal Analysis.

AI Integration:

AI-powered signal analysis for distributed fiber optic sensing. Python simulation of laser propagation and fiber optic signal attenuation.

Free/Open-Source AI Tools: Google Colab, Python, NumPy, SciPy and Scikit-learn/ TensorFlow.

UNIT III: Introduction to Quantum Mechanics & Quantum Computing**10 hrs****Core Content:**

Introduction to quantum mechanics, de-Broglie concept (matter waves), Heisenberg's Uncertainty Principle, Wave function, Postulates of Quantum Mechanics, Schrödinger's time-independent wave equation, Particle in a one-dimensional potential well, Quantum tunnelling (basic idea).

Introduction to Quantum Computing: Classical bits vs Qubits, Concept of superposition and Entanglement, Basic Quantum Operations: Hadamard gate (idea of superposition), Pauli-X gate, Basic idea of quantum measurement. Applications (qualitative only): AI-Assisted Tunnel Diodes and cold emission of electrons, Basic idea of quantum cryptography and secure communication.

AI Integration: AI-Assisted Analysis of Tunnel Diode Characteristics using Quantum Tunnelling Concepts, AI-Assisted Quantum Key Distribution for Secure Communication

Free/Open-Source AI Tools: Google Colab, Python and Scikit-learn.

UNIT IV: Semiconductors and Magnetic Materials**8 hrs****Core Content:**

Semiconductors: Introduction, Fermi energy and Fermi level in intrinsic and extrinsic semiconductors, Expression for concentration of electrons in conduction band and holes in valence band, Electrical conductivity, Expressions for drift and diffusion currents, Hall effect, Expression for Hall coefficient and its applications.

Magnetism: Origin of magnetism, Dia, Para, Ferro, Ferri and Anti-ferro magnetic materials, Hysteresis, Soft and Hard magnetic materials. Applications (qualitative only): Magnetic data storage (MRAM), Hard Disk Drive and Magnetic shielding.

AI Integration: AI-Assisted Hall Effect Analysis for Semiconductor Characterization, AI-Assisted Magnetic Data Storage and MRAM Optimisation

Free/Open-Source AI Tools: Google Colab, Python and Scikit-learn.

UNIT V: Low Temperature Physics and Superconductivity**9hrs****Core Content:**

Introduction: Low temperature and its importance, Properties of materials at low temperatures, Liquefaction of gases (basic concept only), Liquid nitrogen and liquid helium (basic properties and uses). Applications (basic ideas): MRI, Rocket fuels, Cryogenic Cooling of High-Performance Computing System

Superconductors: Definitions, Meissner effect, Type I and Type II superconductors, BCS theory: Cooper pairs and energy gap (qualitative), DC and AC Josephson effect. Applications (qualitative only): SQUIDS, magnetic levitation and lossless power transmission. Superconducting qubits: Charge, Flux, Phase.

AI Integration: AI-Assisted Cryogenic Cooling Optimization for High-Performance Computing Systems, AI-Assisted Stability Control in Superconducting Magnetic Levitation Systems

Free/Open-Source AI Tools: Google Colab, Python and Scikit-learn.

Course Outcomes (COs)

After completion of the course, the student will be able to

CO	Course Outcomes	Bloom's Level
CO1	Apply the principles of interference, diffraction and polarization to analyze optical systems and engineering applications.	L3 (Apply)
CO2	Analyze the working of lasers, fiber optics and sensing systems with basic AI-assisted optical communication concepts.	L4 (Analyze)
CO3	Apply the concepts of quantum mechanics and quantum computing to explain modern computational and secure communication systems.	L3 (Apply)
CO4	Analyse semiconductor characteristics and magnetic material behaviour for engineering and electronic applications.	L4 (Analyze)
CO5	Apply the concepts of low-temperature physics and superconductivity in modern technological applications and intelligent systems.	L3 (Apply)

CO – PO Articulation Matrix

Scale: 3 – Strong, 2 – Moderate, 1 – Slight, – No Correlation

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	1	2	-	-	-	-	-	-
CO2	3	3	2	2	2	-	-	-	-	-	-
CO3	3	2	2	2	2	-	-	-	-	-	-
CO4	3	3	2	2	2	-	-	-	-	-	-
CO5	3	2	1	2	2	-	-	-	-	-	-

Textbooks

1. Avadhanulu, Kshirsagar & Arun Murthy, Engineering Physics, S. Chand, 2019.
2. D. K. Bhattacharya & Poonam Tandon, Engineering Physics, Oxford Press, 2015.

Reference Books

3. B. K. Pandey & S. Chaturvedi, Engineering Physics, Cengage Learning, 2021.
4. S. O. Pillai, Solid State Physics, New Age International.
5. Nielsen & Chuang, Quantum Computation & Quantum Information, Cambridge.
6. Saleh & Teich, Fundamentals of Photonics, Wiley.
7. G. Keiser, Optical Fiber Communications, McGraw Hill.
8. M. Tinkham, Introduction to Superconductivity, Dover Publications.

Web Resources & NPTEL Links

- NPTEL – nptel.ac.in
- MIT OpenCourseWare – ocw.mit.edu
- HyperPhysics – hyperphysics.phy-astr.gsu.edu
- PhET Simulations – phet.colorado.edu
- Library of Congress – loc.gov/rr/scitech/selected-internet/physics.html

Recommended Free & Open-Source AI/Computational Tools

- Python (NumPy, SciPy, OpenCV, Matplotlib) – FREE on Google Colab for wave optics and signal analysis
- IBM Qiskit (free, open-source) – For quantum computing and superconducting qubit simulations on Google Colab
- TensorFlow / Keras (free, open-source) – For photonic and optical pattern recognition
- scikit-learn (free, open-source) – For semiconductor property prediction and LiDAR signal classification
- PhET Interactive Simulations (free) – For wave optics, quantum mechanics and semiconductors

I Year II Semester

Course Code	Title	L	T	P	C
26APH05P	PHYSICS FOR ADVANCED COMPUTING LAB (AI INTEGRATED)	0	0	3	1.5

Prerequisites: Basic Physics and Mathematics (Class XII level); no concurrent prerequisite

Course Objectives

1. To determine fundamental optical properties of materials—including dispersive power, thin-film thickness, radius of curvature, and light wavelengths—using prism, wedge, Newton's rings, and diffraction grating setups.
2. To investigate photon dynamics and laser behavior by measuring laser wavelength through diffraction and estimating Planck's constant via the photoelectric effect.
3. To analyze the electrical, magnetic, and semiconductor characteristics of materials—including B–H hysteresis curves, the Hall effect, energy band gap, resistivity, and thermistor performance—for computing hardware applications.
4. To measure fundamental atomic and thermodynamic quantities, specifically the electron charge-to-mass ratio (e/m) and Stefan's constant, using calibrated experimental setups.
5. To evaluate magnetic field distributions along the axis of a current-carrying coil using standard electromagnetic measuring instruments.

List of Experiments (Any ten experiments need to be performed)

Part A — Regular Experiments

1. Determination of the dispersive power of the prism
2. Determination of the thickness of a thin object by the wedge method
3. Determination of the radius of curvature of the lens by Newton's rings
4. Determination of wavelengths of various colours of mercury spectrum using a diffraction grating in the normal incidence method
5. Laser: Determination of wavelength using a diffraction grating
6. Determination of the numerical aperture of an optical fibre.
7. Estimation of Planck's constant using the photoelectric effect
8. Determination of e/m of an electron by the J.J. Thomson method.
9. Study the variation of B versus H by magnetising the magnetic material (B-H curve)
10. Determination of the resistivity of a semiconductor by the four-probe method.
11. Determination of the energy gap of a semiconductor using a p-n junction diode
12. Determination of Hall voltage and Hall coefficient of a given semiconductor using the Hall effect
13. Determination of Temperature coefficient of a thermistor
14. Determination of Stefan's constant by using incandescence lamp
15. Comparison of thermal insulation – boil-off method.

Part B — AI Integration Lab Exercises

Any two (2) of the following AI exercises must be completed using Python / Google Colab or an equivalent AI/LLM tool.

1. **Optics Data Automation (Python / Matplotlib):** input raw angle-of-deviation data from the prism experiment; fit a polynomial to the deviation-vs-angle curve; extract the angle of minimum deviation and compute the dispersive power; overlay the fitted curve on the experimental data points.
2. **Fringe Analysis using OpenCV (Python / Google Colab):** import Newton's-ring or wedge-fringe images; apply Gaussian blur and Canny edge detection to locate fringe centres; compute fringe diameters automatically; calculate radius of curvature or thickness and compare with manual measurements.
3. **Planck's Constant Estimation (Python / NumPy):** input stopping-potential vs. frequency data from the photoelectric experiment; apply linear regression (V_0 vs. ν); extract the slope to compute h and the y-intercept to find the work function; plot the regression line and quantify percentage error against the accepted value 6.626×10^{-34} J·s.

AI Integration:

AI integration adds two to three exercises after the regular experiments: (i) Python automation of prism deviation curve-fitting and dispersive-power extraction using Matplotlib/NumPy; (ii) OpenCV-based fringe-image analysis for Newton's rings and the wedge method, automating diameter measurement and radius-of-curvature computation; (iii) linear regression on photoelectric-effect data to extract Planck's constant and the work function — reinforcing the connection between the physical experiment and quantum theory, and giving CSE students hands-on practice applying programming and data-analysis skills to experimental physics.

Free / Open-Source AI Tools

- Python (NumPy, SciPy, OpenCV, Matplotlib) on Google Colab (free) — optics automation, fringe analysis, photoelectric regression, and EM simulations
- scikit-learn / TensorFlow (free, open-source) — semiconductor data analysis and signal classification
- PhET Interactive Simulations (free) — wave optics, semiconductors, and quantum phenomena: phet.colorado.edu
- Virtual Labs — IIT (free): https://iitb.vlabs.co.in/discipline.html?discipline=Physical_Sciences — online simulation of optics, semiconductor, and EM experiments
- HyperPhysics (free) — concept reference for optics, quantum, and EM theory: hyperphysics.phy-astr.gsu.edu

Course Outcomes

After successful completion of this lab course, the student will be able to:

CO	Course Outcomes	Bloom's Level
CO1	Compute key optical properties—including dispersive power, thin-film thickness, radius of curvature, and spectral wavelengths—using interference and diffraction techniques.	L3 (Apply)
CO2	Analyze quantum photo-emission dynamics and laser diffraction phenomena to estimate Planck's constant and determine laser wavelengths.	L3–L4 (Apply, Analyse)

CO3	Evaluate electrical, magnetic, and semiconductor properties—such as B–H curves, Hall coefficient, energy band gap, resistivity, and thermistor characteristics—for electronic device applications.	L4 (Analyse)
CO4	Determine fundamental physical and thermodynamic constants, including the electron charge-to-mass ratio (e/m) and Stefan's constant, from empirical data.	L3–L4 (Apply, Analyse)
CO5	Evaluate magnetic field distributions along current-carrying coils and compare axial field behavior against theoretical electromagnetic models.	L5–L6 (Evaluate, Create)

CO – PO Articulation Matrix (NBA)

Scale: 3 – Strong | 2 – Moderate | 1 – Slight | – No Correlation

CO	Bloom's Lvl	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	L3	3	2	2	3	2	–	–	–	–	–	–
CO2	L3–L4	3	3	2	3	3	–	–	–	–	–	–
CO3	L4	3	3	2	3	2	–	–	–	–	–	–
CO4	L3–L4	3	2	2	3	2	–	–	–	–	–	–
CO5	L5–L6	2	3	3	3	3	–	–	–	–	–	–

Recommended Free & Open-Source AI / Computational Tools

Python (NumPy, SciPy, OpenCV, Matplotlib) on Google Colab (free) – optics automation, fringe analysis, photoelectric regression, EM simulations

scikit-learn (free, open-source) – semiconductor data analysis and signal classification

OpenEMS (free, open-source) – electromagnetic field simulation and antenna design

PhET Interactive Simulations (free) – wave optics, semiconductors, and quantum phenomena: phet.colorado.edu

Virtual Labs – IIT (free): https://iitb.vlabs.co.in/discipline.html?discipline=Physical_Sciences – online simulation of optics, semiconductor and EM experiments

HyperPhysics (free) – concept reference for optics, quantum and EM theory: hyperphysics.phy-astr.gsu.edu

LibreOffice Calc (free, open-source) – standardised data recording templates for all experiments

Note: Lab record to be submitted for every experiment at the next session. AI exercises to be submitted as Google Colab (.ipynb) notebooks.

Reference Books

1. M. N. Avadhanulu & P. G. Kshirsagar, A Textbook of Engineering Physics, S. Chand & Company, 2019.
2. H. K. Malik & A. K. Singh, Engineering Physics, Tata McGraw-Hill, 2010.

3. R. K. Gaur & S. L. Gupta, Engineering Physics, Dhanpat Rai Publications, 8th Edition, 2012.
4. B. K. Pandey & S. Chaturvedi, Engineering Physics, Cengage Learning, 2012.
5. D. K. Bhattacharya & Poonam Tandon, Engineering Physics, Oxford University Press, 2015.